

Adwaith B. Uday

PhD Candidate

Department of Anatomy and Cell Biology, McGill University | Montreal, Canada

+1-514-946-8546 | adwaith.biniboseuday@mail.mcgill.ca | [adwaith-b-uday](https://www.linkedin.com/in/adwaith-b-uday) | 0009-0005-1524-6307

Education

McGill University

PhD in Cell Biology

Montreal, Canada

Sep 2022 - present

- **GPA:** 4.0 / 4.0
- **Advisor:** Dr. Natalie Zeytuni, Department of Anatomy and Cell Biology, McGill University, Canada

Indian Institute of Science

Master of Science in Biology

Bangalore, India

Aug 2021 - Jul 2022

- **CGPA:** 8.0/10.0
- **Thesis title:** Position of the initiation factor 3 on empty small ribosomal subunit in *Escherichia coli*
- **Thesis advisor:** Dr. Tanweer Hussain, Department of Molecular Reproduction, Development and Genetics, Indian Institute of Science, Bangalore, India

Indian Institute of Science

Bachelor of Science (Research) in Biology

Bangalore, India

Aug 2017 - Jul 2021

- **CGPA:** 8.0/10.0
- **Thesis title:** Structural basis for the role of the initiation factor 2 in kinetically favoured pathway of 30S pre-initiation complex assembly in *Escherichia coli*
- **Thesis advisor:** Dr. Tanweer Hussain, Department of Molecular Reproduction, Development and Genetics, Indian Institute of Science, Bangalore, India

Research Experience

McGill University

Graduate research assistant (Ph.D. student)

Montreal, Canada

Sep 2022 - present

- Main project focuses on structural characterization of allosteric regulation in bacterial and mitochondrial self-compartmentalized protease systems using cryo-electron microscopy (cryo-EM), resulting in multiple co-first-author publications.
- Work conducted in collaboration with Dr. Siavash Vahidi (University of Guelph), with my contributions including cryo-EM sample preparation, data collection, processing, and structural analysis.

Indian Institute of Science

Graduate research intern

Bangalore, India

Aug 2021 - Apr 2022

- The project focused on obtaining the structure of bacterial small ribosomal subunit (30S) in complex with initiation factor 3 (IF3) using cryo-EM. This complex is one of the first intermediates in prokaryotic translation initiation. There was an uncertainty in the position of IF3 on empty 30S subunit, and our structure cleared this uncertainty.

Indian Institute of Science

Undergraduate research intern

Bangalore, India

Aug 2019 - Jul 2021

- The project focused on understanding the structural basis of the preferred kinetic pathway during the formation of 30S pre-initiation complex in prokaryotic translation.

Drug Discovery Hackathon

Team member

Bangalore, India

Jul 2020 - Apr 2022

- Member of a six-person IISc team participating in the Government of India Drug Discovery Hackathon, applying computational approaches to drug-discovery problem statements.

International Genetically Engineered Machines (iGEM)

Team member

Bangalore, India

May - Oct 2018

- Member of the IISc-Bangalore iGEM 2018 team developing a bacteriophage-based approach to antibiotic resistance, and represented the team at the iGEM Giant Jamboree in Boston.

Awards and Fellowships

- 2026 **Larry Calvert Travel Award**, Canadian National Committee for Crystallography
- 2026 **Best Student Talk**, 52nd Annual Meeting of the Microscopy Society of Canada, Montreal, Canada
- 2025-28 **Doctoral research scholarship**, Fonds de Recherche du Québec - Santé, Quebec, Canada
- 2025 **Prix Annuel d'Excellence**, Centre de Recherche en Biologie Structurale, McGill University, Canada
- 2025 **OMY - Yale Travel Award**, Centre de Recherche en Biologie Structurale, McGill University, Canada

2024	Graduate Research Enhancement and Travel (GREAT) Awards , Department of Anatomy and Cell Biology, McGill University, Canada
2024	David G. Guthrie Fellowship in Medicine , McGill University Graduate and Postdoctoral Studies, McGill University, Canada
2024	Annual studentship , Centre de Recherche en Biologie Structurale, McGill University, Canada
2023	Differential fee waiver award , Centre de Recherche en Biologie Structurale, McGill University, Canada
2021	Winner , Drug discovery hackathon - phase 1, Government of India
2018	Gold medal , International Genetically Engineered Machines (iGEM), iGEM Foundation
2017-22	Fellowship , Kishore Vaigyanik Protsahan Yojana (KVPY) scholarship exam, Department of Science and Technology, Govt. of India

Leadership, Mentorship & Academic Service

McGill Anatomy and Cell Biology Graduate Student Society

McGill University

Co-President

Sep 2025 - present

- Co-led the organization of an off-site departmental research retreat with approximately 90 participants, bringing together graduate students, faculty, staff, and an external keynote speaker. The event was organized entirely by the graduate student society, including scientific program planning and coordination with the venue and catering.
- Coordinated the department's monthly graduate student seminar series, including hosting an invited speaker from Harvard University.
- Represented graduate student interests in discussions with faculty leadership, including meetings with the Associate Dean of the Faculty of Medicine and Health Sciences on issues related to graduate student welfare, funding, and university policies affecting graduate trainees.
- Helped coordinate community-building activities, including the departmental holiday event in collaboration with administrative staff, and additional community-building and networking events for graduate students throughout the year.

McGill Anatomy and Cell Biology Graduate Student Society

McGill University

VP Finance

Sep 2024 - Aug 2025

- Involved in managing the financial affairs of the society.
- Conducted a graduate student survey to assess awareness of university funding policies and identify potential discrepancies in stipend payments.

Student Council, Centre de Recherche en Biologie Structurale

McGill University

VP Academics

Sep 2024 - Aug 2025

- Coordinated student research seminars at the CRBS, helping provide trainees with opportunities to present and discuss their research within the structural biology community.
- Contributed to the organization of the CRBS Annual Symposium through volunteer support.

Centre de Recherche en Biologie Structurale (CRBS) Mentorship program

McGill University

Student mentor

Sep 2024 - Aug 2025

- Volunteered as a mentor in the CRBS mentorship program, supporting junior trainees with questions related to graduate studies, research training, and academic development.
- Mentored three trainees individually—two undergraduate students and one PhD student.

DIALECTICS International Webinar series

Sree Narayana College for Women

Invited talk

Feb 2024

- Delivered an invited outreach talk, "*Photographing the Molecules of Life*," introducing undergraduate students to structural biology and its relevance to biomedical research.
- Presented foundational concepts and techniques in structural biology, including cryo-electron microscopy, with the goal of increasing student exposure to modern structural methods and research careers.

Department of Anatomy and Cell Biology

McGill University

Undergraduate Research Mentor and Supervisor

May 2023 - May 2024

- Mentored and supervised undergraduate student Melina Chen in the lab, providing training in experimental techniques, data analysis, and interpretation.

Green Labs Initiative, McGill Sustainability

McGill University

Lab Representative

Jan 2023 - present

- Serve as the lab's representative for McGill's Green Labs initiative, supporting efforts to reduce single-use plastics by transitioning selected laboratory consumables, including serological pipettes and Petri dishes, to reusable glassware where feasible.
- Coordinate the laboratory's Styrofoam recycling efforts.

McGill Anatomy and Cell Biology Graduate Student Society

McGill University

International Student Representative

Sep 2022 - Aug 2024

- Represented the interests and concerns of international graduate students within the departmental graduate student society and served as a point of contact for issues affecting international trainees.

Publications

- Davis B.T.V.*, **Uday A.B.***, Haris A.*, Keszei A.F.A.*, et al. Resolving hidden stoichiometries in bacterial proteasome activator (Bpa)-substrate complexes by cryo-EM and charge detection mass spectrometry. *J. Am. Chem. Soc.* (2026). doi:10.1021/jacs.6c09027 (*Equal contribution.)
- Goncalves M.M.*, **Uday A.B.***, Watson I.D.G.*, Chau A.M., Kalhor-Monfared S., et al. Allosteric inhibition of human mitochondrial ClpP by hijacking the Dordaviprone (ONC201) binding site. *Nat. Commun.* (2026) (*Equal contribution.) (Accepted)
- Goncalves M.M.*, **Uday A.B.***, Watson I.D.G.*, Chau A.M., et al. A luminal pocket allosterically inhibits human mitochondrial ClpP through conformational strain. *Sci. Adv.* (*Equal contribution.) (Under Review)
- Wang Z.*, **Uday A.B.***, et al. Structural tuning of native type V fimbriae shapes mechanical specialization in *Porphyromonas gingivalis*. (*Equal contribution.) (Under Review)
- Plourde A.*, **Uday A.B.***, Forrester T.J.B., Zeytuni N., Vahidi S. Structural basis of substrate recognition for proteasome degradation by prokaryotic ubiquitin-like protein ligase PafA. *Nat. Commun.* (*Equal contribution.) (Under Revision)
- Goncalves M.M.*, **Uday A.B.***, Forrester T.J.B.*, Currie S.Q.W., Kim A.S., Feng Y., et al. Mechanism of allosteric activation in human mitochondrial ClpP protease. *Proc. Natl. Acad. Sci. U.S.A.* 122(16), e2419881122 (2025). doi:10.1073/pnas.2419881122. (*Equal contribution.)
- Turner M.*, **Uday A.B.***, Velyvis A., Rennella E., Zeytuni N., Vahidi S. Structural basis for allosteric modulation of *M. tuberculosis* proteasome core particle. *Nat. Commun.* 16, 3138 (2025). doi:10.1038/s41467-025-58430-0. (*Equal contribution.)
- **Uday A.B.***, Mishra R.K.*, Hussain T. Initiation factor 3 bound to the 30S ribosomal subunit in an initial step of translation. *Proteins* 93(1), 279-286 (2025). doi:10.1002/prot.26655. (*Equal contribution.)
- Mishra R.K., Sharma P., Khaja F.T., **Uday A.B.**, Hussain T. Atomic structure of wheat ribosome reveals unique features of the plant ribosomes. *Structure* 32(5), 562-574.e3 (2024). doi:10.1016/j.str.2024.02.006.
- Beal J., Farny N.G., Haddock-Angelli T., **iGEM Interlab Study Contributors**, et al. Robust estimation of bacterial cell count from optical density. *Commun. Biol.* 3, 512 (2020). doi:10.1038/s42003-020-01127-5. (Adwaith B. Uday is a member of the iGEM Interlab Study Contributors.)

Selected Presentations

- **Uday A.B.**, Goncalves M.M., Vahidi S., Zeytuni N. *Structural insights into the mechanism of action of novel allosteric inhibitors against the human mitochondrial ClpP protease.* **Poster presentation**, 27th Congress and General Assembly of the International Union of Crystallography, Calgary, Canada (May 2026).
- **Uday A.B.**, Goncalves M.M., Vahidi S., Zeytuni N. *Structural insights into the mechanism of action of novel allosteric inhibitors against the human mitochondrial ClpP protease.* **Oral presentation**, 52nd Annual Meeting of the Microscopy Society of Canada, Montreal, Canada (May 2026).
- **Uday A.B.**, Goncalves M.M., Vahidi S., Zeytuni N. *Structural insights into the mechanism of action of novel allosteric inhibitors against the human mitochondrial ClpP protease.* **Oral presentation**, 7th Annual Symposium, Centre de Recherche en Biologie Structurale, McGill University, Canada (Nov 2025).
- **Uday A.B.**, Turner M., Vahidi S., Zeytuni N. *Structural basis for allosteric regulation of the proteasome core particle.* **Poster presentation**, Yale Biophysics Symposium, New Haven, USA (May 2025).
- **Uday A.B.**, Zeytuni N. *Allosteric regulation in self-compartmentalized proteases.* **Oral presentation**, Research in Progress Seminar Series, Department of Anatomy and Cell Biology, McGill University, Montreal, Canada (Feb 2025).
- **Uday A.B.**, Turner M., Vahidi S., Zeytuni N. *Structural basis for allosteric regulation of the proteasome core particle.* **Poster presentation**, 69th Annual Meeting of the Biophysical Society, Los Angeles, USA (Feb 2025).
- **Uday A.B.**, Turner M., Vahidi S., Zeytuni N. *Allosteric regulation of 20S proteasome core particle from Mycobacterium tuberculosis.* **Poster presentation**, Annual Symposium, Centre de Recherche en Biologie Structurale, McGill University, Canada (Nov 2024).